

Experimental and theoretical study on large amplitude vibrations of clamped rubber plates

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ABSTRACT

In this paper, the large amplitude forced vibrations of thin rectangular plates made of different types of rubbers are investigated both experimentally and theoretically. The excitation is provided by a concentrated transversal harmonic load. Clamped boundary conditions at the edges are considered, while rotary inertia, geometric imperfections and shear deformation are neglected since they are negligible for the studied cases. The von Kármán nonlinear strain-displacement relationships are used in the theoretical study; the viscoelastic behaviour of the material is modelled using the Kelvin-Voigt model, which introduces nonlinear damping. An equivalent viscous damping model has also been created for comparison. In-plane pre-loads applied during the assembly of the plate to the frame are taken into account. In the experimental study, two rubber plates with different material and thicknesses have been considered; a silicone plate and a neoprene plate. The plates have been fixed to a heavy rectangular metal frame with an initial stretching. The large amplitude vibrations of the plates in the spectral neighbourhood of the first resonance have been measured at various harmonic force levels. A laser Doppler vibrometer has been used to measure the plate response. Maximum vibration amplitude larger than three times the thickness of the plate has been achieved, corresponding to a hardening type nonlinear response. Experimental frequency-response curves have been very satisfactorily compared to numerical results. Results show that the identified retardation time increases when the excitation level is increased, similar to the equivalent viscous damping but to a lesser extent due to its nonlinear nature. The nonlinearity introduced by the Kelvin-Voigt viscoelasticity model is found to be not sufficient to capture the dissipation present in the rubber plates during large amplitude vibrations.

1. Introduction

Plates are widely used structural elements, which are flat and with a relatively small thickness compared to the in-plane dimensions. They exhibit geometrically nonlinear vibrations under certain loading conditions that have been studied over the past few decades by many scholars. Detailed review of these studies can be found in the literature [1–3]. Many scholars have studied such problems generating a vast literature in terms of both theoretical and experimental front in the elastic plate domain. The study on viscoelastic materials is found to be very limited due to the complexities in modeling the time dependent stress-strain responses. Metals under high temperature, elastomer and many biological materials show viscoelastic characteristics. Nowadays, elastomers are being used increasingly in many automobile and aerospace applications for their excellent damping characteristics. Moreover, they are also used in combination with composite or metal layers

to produce effective structural members. Thus, modeling their dynamic response is crucial for effective design in such application areas. Viscoelastic material responses can be modelled using many different models; among them Maxwell, Kelvin-Voigt, standard linear solid and Boltzmann models are commonly used [4,5]. Stress relaxation and creep are unique properties of these materials and are captured well in standard linear solid and Boltzmann models. Even though Kelvin-Voigt model has limitations in describing relaxation of stress, it accurately captures creep phenomenon and is widely used for modeling vibrations of viscoelastic materials. Mahmoodi et al. [6] and Ghayesh et al. [7] investigated theoretically and experimentally the nonlinear vibration response of viscoelastic beams by using the Kelvin-Voigt model for the cantilever and the simply supported and axially travelling conditions, respectively. Viscoelastic plates have also been studied by using different models. Notably, Esmailzadeh and Jalali [8] and Sun and Zhang [9] used the Kelvin-Voigt model to study the nonlinear vibrations of

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plates, while the experimental investigation of sandwich composite plates with viscoelastic layers was carried out by Balkan and Mecitoglu [10]. Amabili [11] compared the viscoelastic plates to its equivalent viscous model in order to understand the effects of nonlinear damping terms in the forced vibration response. Nonlinear vibration analysis of viscoelastic panels, described by Kelvin-Voigt model and using a very simple model with only two degrees of freedom, was conducted by Han and Hu [12]. Standard linear solid type model was used to represent the viscoelastic characteristics in nonlinear vibrations and dynamic stability of plates by Eshmatov [13] and Salehi and Safi-Djahanshahi [14]. Kim and Kim [15] used the Boltzmann model to study the nonlinear vibration of viscoelastic laminated composite plates, and concluded that geometric nonlinearities do not affect the dissipative characteristics. Though analytical studies on vibration (linear and nonlinear) of viscoelastic plates using different models are available, experimental results are quite scarce in the literature. In this paper, the geometrically nonlinear response of clamped rubber plates is experimentally studied, and the results are compared to those obtained by theoretical nonlinear models by using (i) the Kelvin-Voigt viscoelasticity and (ii) a viscous damping model.

2. Experimental set-up

The experimental setup is composed of two subsystems: (i) the vibrating rubber plate with its supporting frame, and (ii) the transducers and data acquisition system used to perform forced vibration testing. In the following subsections these two subsystems will be described briefly.

2.1. The rubber plates

Two viscoelastic plates are considered here; a silicone plate with 1.5 mm thickness and a neoprene plate with 3 mm thickness. The dimensions of both plates are 0.26×0.26 m and their nonlinear vibrations were investigated. In reality, bigger square plates were cut from commercially available rolls, as part of the area had to be used to realize the desired boundary conditions. The silicone plate has the hardness of 50 A and density of 1430 kg/m^3 as per commercial catalogue, whereas neoprene plate has 70 A hardness and 1230 kg/m^3 density. While part of this study is intended to obtain the material characterization of these plates, it has to be noted that the Poisson's ratio was considered to be 0.5 (incompressibility condition).

As both materials are relatively soft and yielding, an in-plane stretching (load in both directions) was adopted to guarantee the flatness of the plates. In-plane tension was obtained by cutting 28 holes in the plate and mounting it onto 28 perpendicular M12 fasteners, installed on a metal frame. The holes in the plate were uniformly distributed on a square perimeter, the side of which was 0.31 m. The M12 rods, again equally-spaced, were cantered onto a square perimeter, the side of which was 0.315 m. The plates were therefore manually stretched to be mounted onto the threaded rods. The diameter of the holes is slightly bigger than the diameter of the bolts. The resulting in-plane tension can be described as an initial perpendicular displacement of 0.0025 m imposed to each side of the 0.31×0.31 m square. As the distance between the holes is modest, the stretch can be considered uniform along the length of the square. A representation of the pre-stretching procedure is shown in Fig. 1. This initial in-plane stretch should be perfectly symmetric; however, dynamic tests, described later in this paper, reveal that the tension state is not perfectly symmetric. This can be due to the localized deformation of the plates at the anchor points. Moreover, few tenths of a millimeter of stretch correspond to a large variation in the stress state. The square shape introduces even higher uncertainty about the strain distribution, as its corners constitute stress concentration factors. As mentioned above, the vibrating portion of the plates considered here is a square area, 0.26 m of side, concentric with respect to the previously described offsets of holes and threaded rods. The 0.26 m side of the vibrating square area is defined in the

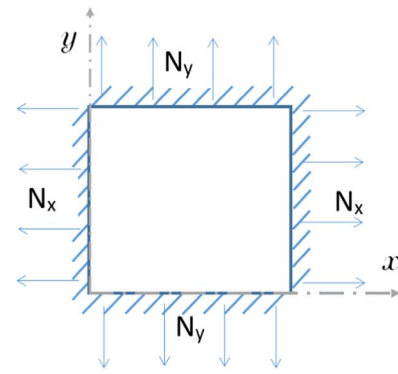


Fig. 1. Clamped viscoelastic plate with in-plane loads.

stretched side, and therefore corresponds to a smaller area in the relaxed state. Fixed boundary conditions were realized pressing the rubber plates between two heavy-weight metal frames matching square openings leaving the vibrating area free. The thickness, width and rigidity of the frames guarantee a clamped condition. The resulting system is then connected rigidly to the ground. The two metal frames exert a pressure on the rubber plates by means of the same threaded elements used to keep the rubber into a pre-stretched state. The resulting friction force between metal and viscoelastic plate is important in constituting the boundary condition; however, the two materials have a low friction coefficient and would require very a high torque on the threaded bolts. As rubber is nearly incompressible, higher torques would squeeze it between the two rigid frames, forcing it eventually to move into the free area. This complex transfer of material would buckle the originally free flat area and would appear as a pre-stretch reduction. In order to avoid this, sandpaper was glued onto a portion of the metal-silicone/neoprene contact area to increase the friction coefficient. Fig. 2 shows the plates clamped using the metal frames.

The planar square vibrating area was maintained vertical during the experiments, with two sides parallel to the horizontal, since the effect of gravity was considered negligible with respect to the pre-stretch. A right-hand coordinate system was considered as having the x direction parallel to the horizontal and the z direction perpendicular to the plate. The positive direction being the face pointing towards the laser from the observer's point of view. With this convention, the y axis will be vertical and pointing upwards. Bottom left corner was taken as the origin for the described coordinate system. Fig. 2 represents the coordinate system adopted, the position of the excitation (E) and measurement (M). A transversal excitation (perpendicular to the plate) was positioned at $x = 0.22$ m and $y = 0.24$ m for the neoprene plate and at $x = 0.11$ m and $y = 0.19$ m for the silicone plate. This location was chosen since modes with both odd and even symmetry had to be excited. The distance from the edges of the domain under study (0.26×0.26 m square perimeter) was chosen as a trade-off. A reduced distance would correspond to a low energy transfer to the structure, being the edges subjected to a fixed boundary condition. A larger distance would instead give a wider unwanted interaction between the vibration of the structure and the excitation system, typical of nonlinear vibrations. A punctual excitation was a rigid aluminium base, 5 mm in diameter and negligible in weight, was centered on the chosen excitation point and glued by means of cyanoacrylic glue. The bond of aluminium and silicone through cyanoacrylic glue is weak, however it was deemed acceptable, the force amplitude involved in these experiments being low.

Fig. 3(a,b) show the stress-strain behaviour of the silicone and neoprene rubbers, respectively, as obtained by uniaxial traction tests with weights on rubber strips. The results show that, even if the tests confirm the hyperelastic rubber behaviour, the stress-strain relationship is practically linear for strains up to the order of 2%, as those achieved during our vibration tests. For this reason, linear visco-elasticity is assumed for the rubbers in the present study, with Young's modulus of

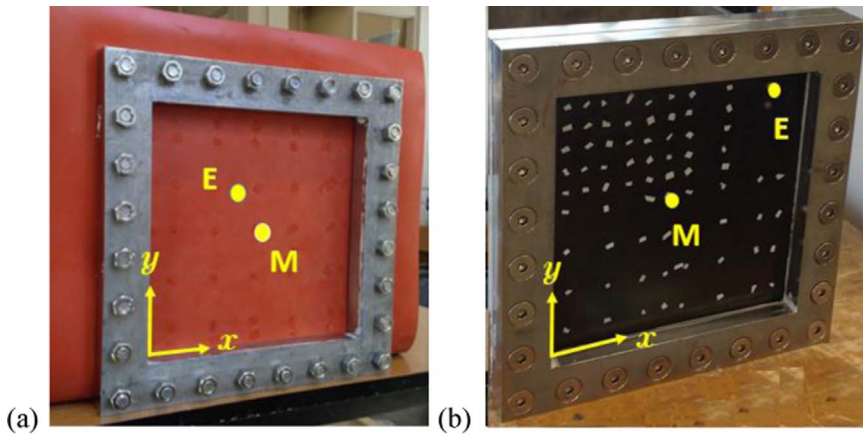


Fig. 2. Viscoelastic plates with their respective co-ordinate system; (a) Silicone plate (b) Neoprene Plate; *E* - Excitation Point; *M* - Measurement Point.

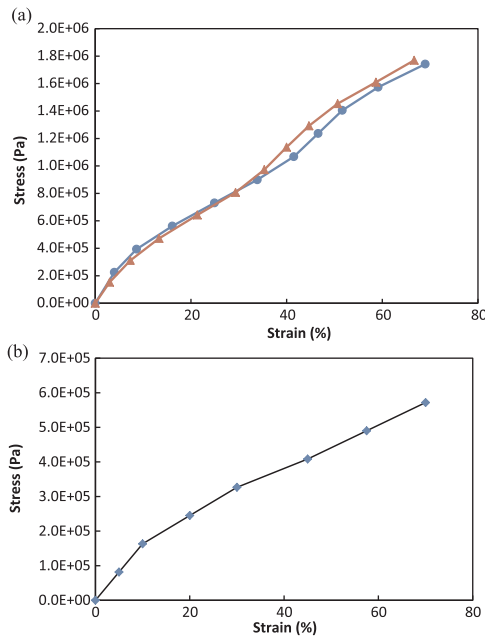


Fig. 3. Uni-axial traction tests on rubbers. (a) Silicone rubber; (b) neoprene rubber.

5.62 MPa for the silicone plate and 2.45 MPa for the neoprene plate. The storage modulus of both silicone and neoprene rubbers were measured around the fundamental frequency of respective plates and found to be 4.28 and 2.62 times the static Young's modulus. This has been taken into account in the model.

2.2. Measurement setup

During dynamic testing, a Brüel Kjær 8203 miniaturized force transducer was used to measure the value of the excitation force. The transducer is connected by a stinger to the Brüel Kjær 4810 electro-dynamic exciter (shaker), powered by a Brüel Kjær 2718 power amplifier. Fig. 4 shows an overall image of the excitation system applied to the silicone/neoprene plate. The shape of normal vibration modes was obtained by means of scanning laser Doppler vibrometer (Polytec PSV-400). This technique allows the non-contact measurement of a multitude of points on the surface of the silicone plate, disposed on a fine grid. A Polytec OFV-505 single point laser Doppler vibrometer was instead used in case of nonlinear vibrations. As the relevant stepped-sine procedure studies nonlinear vibrations around one underlying natural frequency and one vibration mode only, the measurement of one point is sufficient. Non-contact measurement systems were favoured also because they do not introduce unwanted added masses.

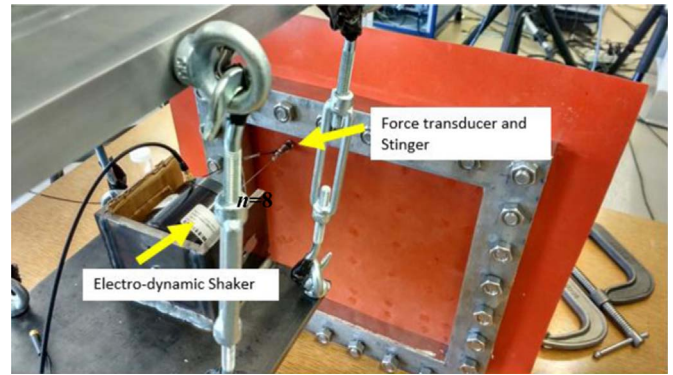


Fig. 4. Excitation setup with electrodynamic shaker.

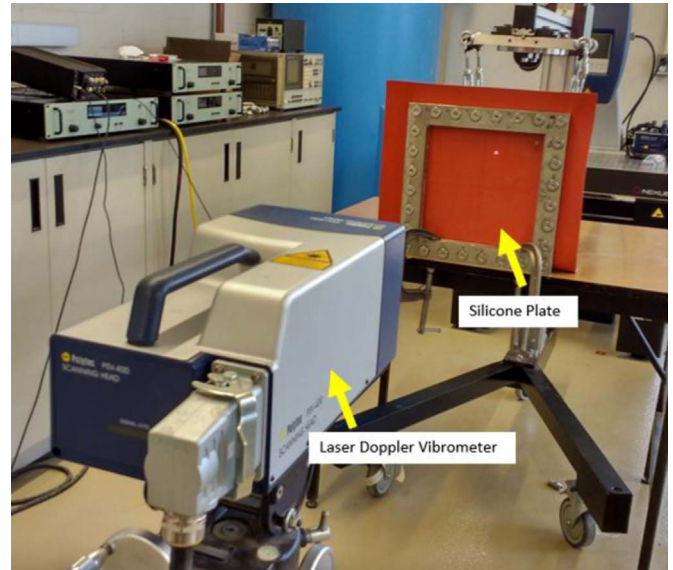


Fig. 5. Measurement setup with Laser Doppler Vibrometer.

Fig. 5 shows the measurement set-up with the sensors described above. Modal analysis was performed using a dedicated Polytec OFV-5000 data acquisition system. Subsequently, the data were transferred and processed by the LMS Test. Lab modal analysis module software. The nonlinear tests required the connection of the laser Doppler vibrometer and of the force transducer to a LMS SCADAS III data acquisition system and the use of the software LMS Test. Lab stepped sine module. Stepped-sine tests were performed at various fixed levels of force amplitude, applying a sinusoidal excitation sweeping monotonously

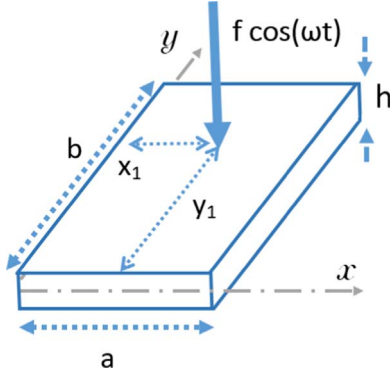


Fig. 6. The viscoelastic plate.

(forwards and then backwards) by small steps the frequency neighbourhood of one chosen natural frequency. The apparatus is capable of maintaining constant amplitude of the sinusoidal excitation by means of a feedback loop control; it must be observed that the control of the input signal to the power amplifier is not an acceptable technique, since the impedance of the structure near resonance changes significantly. The LMS Test. Lab system records the signals from vibration and force sensors both in time and frequency domain.

3. Theoretical model

A viscoelastic rectangular plate with coordinate system $(O; x; y; z)$ having the origin (O) at one corner is considered, as shown in Fig. 6. The plate dimensions in x and y directions are a and b , respectively, and the thickness is h . The displacements of an arbitrary point of coordinates (x, y) on the middle surface of the plate are denoted by u, v and w in the x, y and z directions, respectively. Rotary inertia, shear deformation and geometric imperfections are neglected since the plates tested in the experiments are very thin and are assembled with initial stretching, which ensures an initial flatness of the plate. Clamped boundary condition on all four edges and the von Kármán strain-displacement relationships are considered. According to the von Kármán strain-displacement relationship, the strain components $\epsilon_x, \epsilon_y, \gamma_{xy}$ at an arbitrary point of the plate are related to the middle surface strains $\epsilon_{x,0}, \epsilon_{y,0}, \gamma_{xy,0}$ and to the changes of curvature and torsion of the middle surface k_x, k_y, k_{xy} by

$$\epsilon_x = \epsilon_{x,0} + z k_x, \quad \epsilon_y = \epsilon_{y,0} + z k_y, \quad \gamma_{xy} = \gamma_{xy,0} + z k_{xy}, \quad (1a-c)$$

$$\epsilon_{x,0} = \frac{\partial u}{\partial x} + \frac{1}{2} \left(\frac{\partial w}{\partial x} \right)^2, \quad \epsilon_{y,0} = \frac{\partial v}{\partial y} + \frac{1}{2} \left(\frac{\partial w}{\partial y} \right)^2, \quad \gamma_{xy,0} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y}, \quad (2a-c)$$

$$k_x = -\frac{\partial^2 w}{\partial x^2}, \quad k_y = -\frac{\partial^2 w}{\partial y^2}, \quad k_{xy} = -2 \frac{\partial^2 w}{\partial x \partial y}, \quad (3a-c)$$

The kinetic energy of the plate is given by

$$T = \frac{1}{2} \rho h \int_0^a \int_0^b (\dot{u}^2 + \dot{v}^2 + \dot{w}^2) dx dy, \quad (4)$$

where ρ is the density in kg/m^3 and the over-dot indicates the time derivative. The viscoelastic behaviour of the material is modelled by using the Kelvin-Voigt model. Hence, the stress-strain relationship is defined by

$$\sigma = E(\epsilon + \eta \dot{\epsilon}), \quad (5)$$

that written in extended form is

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} \frac{E}{1-\nu^2} & \frac{\nu E}{1-\nu^2} & 0 \\ \frac{\nu E}{1-\nu^2} & \frac{E}{1-\nu^2} & 0 \\ 0 & 0 & \frac{E}{2(1+\nu)} \end{bmatrix} \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix} + \eta \begin{bmatrix} \frac{E}{1-\nu^2} & \frac{\nu E}{1-\nu^2} & 0 \\ \frac{\nu E}{1-\nu^2} & \frac{E}{1-\nu^2} & 0 \\ 0 & 0 & \frac{E}{2(1+\nu)} \end{bmatrix} \begin{bmatrix} \dot{\epsilon}_x \\ \dot{\epsilon}_y \\ \dot{\gamma}_{xy} \end{bmatrix}, \quad (6)$$

where E is Young's Modulus and η is retardation time in seconds; here the stresses are due to the plate vibration and the contribution of initial pre-load of the plate is not included. In Eq. (6) a linear viscoelastic material is assumed, i.e. linear elasticity is assumed for static loads; the reason for this choice is justified in Section 2.1 where the hyperelastic behaviour of rubber is characterized and the range of strains during experimental tests is discussed. Since it is assumed that the transverse stresses are negligible, the potential energy of the plate is given by

$$U_P = \frac{1}{2} \int_0^{\frac{h}{2}} \int_0^a \int_0^b (\sigma_x \epsilon_x + \sigma_y \epsilon_y + \tau_{xy} \gamma_{xy}) dx dy dz. \quad (7)$$

Using constitutive Eq. (6) and strain-displacements relationships (1), the potential energy of the plate can be written as a sum of the elastic (U_E) and the viscoelastic terms (U_V)

$$U_P = U_E + U_V, \quad (8)$$

where

$$U_E = \frac{Eh}{2(1-\nu^2)} \int_0^a \int_0^b \left(\epsilon_{x,0}^2 + \epsilon_{y,0}^2 + 2\nu \epsilon_{x,0} \epsilon_{y,0} + \frac{1-\nu}{2} \gamma_{xy,0}^2 \right) dx dy + \frac{Eh^3}{2(12(1-\nu^2))} \int_0^a \int_0^b \left(k_x^2 + k_y^2 + 2\nu k_x k_y + \frac{1-\nu}{2} k_{xy}^2 \right) dx dy + O(h^4), \quad (9)$$

$$U_V = \eta \frac{Eh}{(1-\nu^2)} \int_0^a \int_0^b \left(\epsilon_{x,0} \frac{\partial \epsilon_{x,0}}{\partial t} + \epsilon_{y,0} \frac{\partial \epsilon_{y,0}}{\partial t} + \nu \epsilon_{x,0} \frac{\partial \epsilon_{y,0}}{\partial t} + \nu \epsilon_{y,0} \frac{\partial \epsilon_{x,0}}{\partial t} + \frac{1-\nu}{2} \gamma_{xy,0} \frac{\partial \gamma_{xy,0}}{\partial t} \right) dx dy + \frac{Eh^3}{12(1-\nu^2)} \int_0^a \int_0^b \left(k_x \frac{\partial k_x}{\partial t} + k_y \frac{\partial k_y}{\partial t} + \nu k_x \frac{\partial k_y}{\partial t} + \nu k_y \frac{\partial k_x}{\partial t} + \frac{1-\nu}{2} k_{xy} \frac{\partial k_{xy}}{\partial t} \right) dx dy + O(h^4). \quad (10)$$

In-plane loads have been applied to the plate as shown in Fig. 1. The energy from the in-plane loads applied to the plate is given by (see Appendix A)

$$U_M = N_x^s \int_0^a \int_0^b \left(\frac{\partial u}{\partial x} + \left(\frac{\partial w}{\partial y} \right)^2 \right) dx dy + N_y^s \int_0^a \int_0^b \left(\frac{\partial v}{\partial x} + \left(\frac{\partial w}{\partial y} \right)^2 \right) dx dy + \eta \left(N_x^s \int_0^a \int_0^b \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial x} + \left(\frac{\partial w}{\partial y} \right)^2 \right) dx dy + N_y^s \int_0^a \int_0^b \frac{\partial}{\partial t} \left(\frac{\partial v}{\partial x} + \left(\frac{\partial w}{\partial y} \right)^2 \right) dx dy \right), \quad (11)$$

where N_x^s and N_y^s are in-plane loads per unit length in x and y directions, respectively, measured in N/m . Tensile and compressive loads are positive and negative, respectively.

The virtual work done by external transversal punctual harmonic load is given by

$$W = \int_0^a \int_0^b f \cos(\omega t) \delta(x - x_1) \delta(y - y_1) w dx dy = f \cos(\omega t) (w)_{x=x_1, y=y_1}, \quad (12)$$

where f is the harmonic point load (N), applied at the point $(x=x_1, y=y_1)$, and δ is Dirac delta function.

The boundary conditions of the considered plate are given as below,

$$u = v = w = 0, \quad M_x = \pm k \frac{\partial w}{\partial x}, \quad \text{at } x=0, a; \quad (13)$$

$$u = v = w = 0, \quad M_y = \pm k \frac{\partial w}{\partial y}, \quad \text{at } y=0, b; \quad (14)$$

the minus sign in Eqs. (13) and (14) applies at the boundaries $x=0$ and $y=0$. They give any rotational constraint from zero bending moment ($M_x = 0$ and $M_y = 0$, unconstrained rotation, obtained for $k=0$) to perfectly clamped plate ($\partial w / \partial x = 0$ and $\partial w / \partial y = 0$, obtained as limit for $k=\infty$), according to the value of k , where k is the stiffness per unit length of the elastic, distributed rotational springs placed at the four edges. In the present study, a very large value of k is assumed in order to approximate clamped boundary conditions.

In order to reduce the system to finite dimensions, the mid surface displacements u , v and w are expanded by using approximate functions, which satisfy the geometric boundary conditions (13) and (14)

$$\begin{aligned} u(x, y, t) &= \sum_{m=1}^M \sum_{n=1}^N u_{m,n}(t) \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right); \\ v(x, y, t) &= \sum_{m=1}^M \sum_{n=1}^N v_{m,n}(t) \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right); \\ w(x, y, t) &= \sum_{m=1}^M \sum_{n=1}^N w_{m,n}(t) \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right); \end{aligned} \quad (15)$$

where, m and n are the number of half-waves in x and y directions respectively and t is the time. M and N the number of terms used to expand the corresponding displacements. The energy due to the rotational springs is given by

$$\begin{aligned} U_R &= \frac{1}{2} \int_0^b k \left[\left(\left(\frac{\partial w}{\partial x} \right)_{x=0} \right)^2 + \left(\left(\frac{\partial w}{\partial x} \right)_{x=a} \right)^2 \right] dy \\ &+ \frac{1}{2} \int_0^a k \left[\left(\left(\frac{\partial w}{\partial y} \right)_{y=0} \right)^2 + \left(\left(\frac{\partial w}{\partial y} \right)_{y=b} \right)^2 \right] dx. \end{aligned} \quad (16)$$

The following notation is introduced for brevity

$$\mathbf{q} = \{\mathbf{u}_{m,n}, \mathbf{v}_{m,n}, \mathbf{w}_{m,n}\}^T. \quad (17)$$

The generic element of the time-dependent vector of the generalized coordinates $\mathbf{q}$ is referred to as q_j ; the dimension of $\mathbf{q}$ is equal to the degrees of freedom used in the mode expansion, N_q . The generalized forces Q_j are obtained by differentiating the virtual work done by external forces

$$Q_j = \frac{\partial W}{\partial q_j}, \quad \text{for } j=1, \dots, N_q. \quad (18)$$

The Lagrangian equations of motion are given by

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial U}{\partial q_j} = Q_j, \quad \text{for } j=1, \dots, N_q, \quad (19)$$

where, $U=U_p + U_K + U_M$. Eq. (19) can be written in the following matrix form,

$$\mathbf{M}\ddot{\mathbf{q}} + [\mathbf{G} + \mathbf{G}_2(q) + \mathbf{G}_3(q, q)]\dot{\mathbf{q}} + [\mathbf{K} + \mathbf{K}_2(q) + \mathbf{K}_3(q, q)]\mathbf{q} = \mathbf{f}_0 \cos(\omega t) \quad (20)$$

where $\mathbf{M}$ is the diagonal mass matrix of dimension $N_q \times N_q$; $\mathbf{G}$ is the linear viscoelasticity matrix, $\mathbf{G}_2$ gives the quadratic nonlinear viscoelasticity terms, $\mathbf{G}_3$ denotes the cubic nonlinear viscoelasticity terms; $\mathbf{K}$ is the linear stiffness matrix, $\mathbf{K}_2$ gives the quadratic nonlinear stiffness terms, $\mathbf{K}_3$ denotes the cubic nonlinear stiffness terms and $\mathbf{f}_0$ is the vector representing the projection of the concentrated harmonic force on the generalized coordinates. In order to obtain the equations of motion in a suitable form for numerical implementation, the system is multiplied by the inverse of the mass matrix and then is written in the state-space form as follows,

$$\begin{aligned} \dot{\mathbf{q}} = \mathbf{y}; \dot{\mathbf{y}} = & -[\mathbf{M}^{-1}\mathbf{G} + \mathbf{M}^{-1}\mathbf{G}_2(q) + \mathbf{M}^{-1}\mathbf{G}_3(q, q)]\dot{\mathbf{q}} \\ & + [\mathbf{M}^{-1}\mathbf{K} + \mathbf{M}^{-1}\mathbf{K}_2(q) + \mathbf{M}^{-1}\mathbf{K}_3(q, q)]\mathbf{q} + \mathbf{M}^{-1}\mathbf{f}_0 \cos(\omega t); \end{aligned} \quad (21)$$

Eq. (20) can be modified by removing the viscoelasticity terms ($\mathbf{G}, \mathbf{G}_2, \mathbf{G}_3$) and adding the $\mathbf{C}$ viscous damping matrix to generate an equivalent viscous damping model.

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + [\mathbf{K} + \mathbf{K}_2(q) + \mathbf{K}_3(q, q)]\mathbf{q} = \mathbf{f}_0 \cos(\omega t) \quad (22)$$

The state space form of the Eq. (22) is as given below,

$$\begin{aligned} \dot{\mathbf{q}} = \mathbf{y}; \dot{\mathbf{y}} = & -\mathbf{M}^{-1}\mathbf{C}\dot{\mathbf{q}} + [\mathbf{M}^{-1}\mathbf{K} + \mathbf{M}^{-1}\mathbf{K}_2(q) + \mathbf{M}^{-1}\mathbf{K}_3(q, q)]\mathbf{q} \\ & + \mathbf{M}^{-1}\mathbf{f}_0 \cos(\omega t); \end{aligned} \quad (23)$$

The viscous damping matrix is assumed to be given by

$$\mathbf{M}^{-1}\mathbf{C} = \begin{bmatrix} 2\omega_{1,1}\zeta_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 2\omega_{1,1}\zeta_{1,1} \end{bmatrix}, \quad (24)$$

where, $\omega_{1,1}$ is the natural frequency of the mode ($m=1, n=1$), being m and n the number of axial half-waves in the x and y direction respectively, and $\zeta_{1,1}$ is the corresponding assumed modal damping ratio.

4. Numerical simulation

The software Mathematica [16] has been used to perform the surface integrals and to obtain the $2 \times N_q$ first-order ordinary differential equations (ODEs) written in state space form, as shown in Eqs. (21) and (23). They are very long expressions containing nonlinear stiffness and damping terms and are not given in this paper for sake of brevity. Nondimensionalization of variables has been carried out for computational convenience; the vibration amplitude has been divided by the plate thickness and time has been multiplied with the resonance frequency in rad/s. These $2 \times N_q$ first-order ODEs are solved by using a bifurcation analysis software AUTO and also integrated in time. The AUTO software uses pseudo arc-length continuation and collocation methods to follow the solution path. It is important to note that, along with stable solutions, unstable solutions are also obtained [17]. In particular, the analysis has been started by considering the excitation force as the bifurcation parameter, having fixed the excitation frequency far away from the resonance. The solution starts at zero magnitude of excitation and is increased slowly to reach the desired magnitude. Then, the solution is continued by varying the previously fixed excitation frequency, spanning the frequency range around the resonance in order to obtain the frequency-response curves. The equations are also directly integrated in time by using the Adams-Gear scheme to get the time-response of the structure.

5. Comparison and discussion

5.1. Linear analysis

Initially, a linear modal analysis of both the plates has been carried out by exciting them with pseudo-random signals using the measurement apparatus and setup explained earlier. Many Frequency Response Functions (FRF) have been measured on a predefined grid of points to capture the first (lower frequency) few modes of interest. A summation of all the FRFs measured for each plate is shown in Fig. 7. The peaks of the FRFs' sum are marked with their corresponding mode shapes in the format of number of half waves in (x, y) direction. Subsequently, a linear analysis has been performed on the respective nonlinear numerical model by eliminating the nonlinear terms from Eq. (20). The rotational spring constant (k) value has been calculated by using convergence analysis and a value of $k=1000$ N/rad has been found to give good results. Any further increase in the rotational spring constant does not affect significantly the natural frequency of the viscoelastic plate.

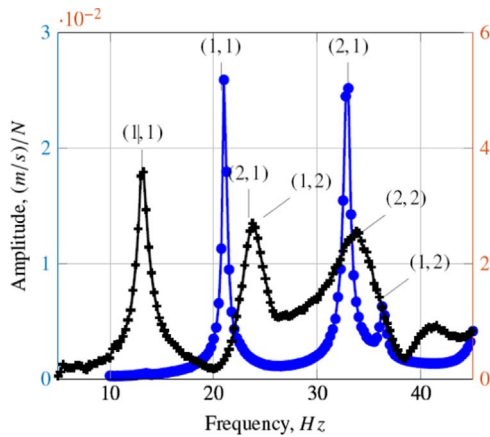


Fig. 7. Sum of Frequency Response Functions; +, neoprene plate (vertical axis scale on the left); *, silicone plate (vertical axis scale on the right).

Further, unknown in-plane loads (N_x^s and N_y^s) applied on the plates during experimental setup have been obtained by an optimization algorithm. The experimentally identified natural frequencies of both plates are matched using a commercial finite element model while adjusting in-plane loads. The optimization criterion is to minimize the error function E_f in the following form:

$$E_f = \sum_{k=1}^4 p_k (f_k^{EXP} - f_k^{COM})^2. \quad (25)$$

In Eq. (25), f_k^{EXP} are the natural frequencies determined experimentally and f_k^{COM} are the frequencies computed from the finite element model used; p_k is a weight coefficient assigned to each of four modes considered. The error function (25) is minimized and the optimum in-plane loads are identified for both plates. These in-plane loads are also obtained by using a linear model with 507 degrees of freedom (DOFs), which is obtained by the expansion (15) taken with $M = N = 13$, and by the Abaqus commercial Finite Element code.

In the nonlinear analysis, reduced-order numerical models with 39 and 54 DOFs, which are the number of terms inserted in the expansion of u , v and w given in Eq. (15), are used. A 39 DOF model with the following generalized coordinates $w_{1,1}, w_{3,1}, w_{5,1}, w_{7,1}, w_{1,3}, w_{3,3}, w_{5,3}, w_{7,3}, w_{1,5}, w_{3,5}, w_{5,5}, w_{7,5}, w_{1,7}, w_{3,7}, w_{5,7}, w_{7,7}, u_{2,1}, u_{4,1}, u_{6,1}, u_{8,1}, u_{2,3}, u_{4,3}, u_{6,3}, u_{8,3}, u_{2,5}, u_{4,5}, u_{6,5}, u_{8,5}, u_{2,7}, u_{4,7}, v_{1,2}, v_{3,2}, v_{5,2}, v_{7,2}, v_{1,4}, v_{3,4}, v_{5,4}, v_{7,4}, v_{1,6}, v_{3,6}, v_{5,6}, v_{7,6}, v_{1,8}, v_{3,8}$ has been built to study the silicone plate. Since, for the neoprene plate the mode (2,1) gives a relevant contribution, a larger 54 DOF model has been developed with the following generalized coordinates $w_{1,1}, w_{1,3}, w_{1,5}, w_{1,7}, w_{3,1}, w_{3,3}, w_{3,5}, w_{3,7}, w_{5,1}, w_{5,3}, w_{5,5}, w_{5,7}, w_{7,1}, w_{7,3}, w_{7,5}, w_{7,7}, u_{2,1}, u_{2,3}, u_{2,5}, u_{2,7}, u_{4,1}, u_{4,3}, u_{4,5}, u_{4,7}, u_{6,1}, u_{6,3}, u_{6,5}, u_{6,7}, u_{8,1}, u_{8,3}, u_{8,5}, u_{8,7}, v_{1,2}, v_{1,4}, v_{1,6}, v_{1,8}, v_{3,2}, v_{3,4}, v_{3,6}, v_{3,8}, v_{5,2}, v_{5,4}, v_{5,6}, v_{5,8}, v_{7,2}, v_{7,4}, v_{7,6}, v_{7,8}$.

It must be observed here that in the nonlinear models (with 39 or 54 DOFs), the in-plane load found to have the best match with the experimental natural frequencies are about 10–20% smaller than the

actual in-plane loads determined by the Finite Element Method (FEM) or by the 507 DOFs linear model. In fact, the reduced number of in-plane modes in the nonlinear model makes it stiffer to in-plane loads. So, in order to keep a small size of the nonlinear model, it is necessary to use different in-plane load values. Table 1 shows the linear natural frequencies along with the in-plane loads used in the nonlinear and linear models.

The first two mode shapes obtained experimentally, by the 507 DOFs linear model and by the commercial FEM code are shown in Table 2 for the neoprene plate and they are very close each other.

5.2. Nonlinear analysis

5.2.1. Comparison with the Kelvin-Voigt model

The nonlinear response of both plates has been measured at various force levels. For the silicone plate, the response has been measured at the centre of the plate (0.13 m, 0.13 m) with the excitation located at (0.11 m, 0.19 m), for four different force levels: 0.01N, 0.04N, 0.07N and 0.1N. The frequency of excitation has been varied from 17 to 25 Hz for sweep-up curves and from 25 to 17 Hz for the sweep-down curves. Sweeping up and down curves are necessary to capture the jump-up and jump-down phenomenon (hysteresis) present in the nonlinear responses. Since, the measurement has been taken at the plate's centre, the influence of the first mode is predominant while the contribution of the subsequent modes is negligible.

The comparison between the experimental results and numerical simulation is shown in Fig. 8. The numerical response obtained by using the software AUTO is shown here and it has been calculated for all the four force levels by using the 39 DOF model. The response of the plate is practically linear for the force level 0.01 N. Numerical results are matching very well with the experimental responses. A retardation time η of 0.000121 s has been used in the numerical model to match the experimental results at the first force level. The response from the plate becomes nonlinear in the consecutive force levels and maximum vibration amplitude of more than 3 times the thickness of the plate has been reached for the largest force level. At the force levels of 0.04 and 0.07 N, the numerical model is accurately capturing the hardening behaviour of the plate. Whereas, at the highest force level, the numerical model slightly over-predicts the nonlinear behaviour of the plate. This could be due to the reduced number of DOFs used in the model. The retardation time of the plate, joined to the viscoelastic dissipation, increases when the vibration amplitudes become larger. In particular, the retardation time η used for matching the experimental results of the second, third and fourth excitation level are 0.000121, 0.000162 and 0.000145 s, respectively.

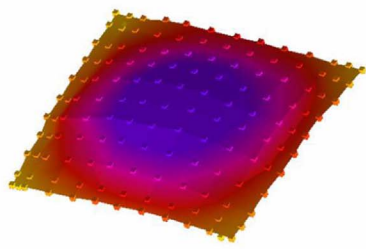
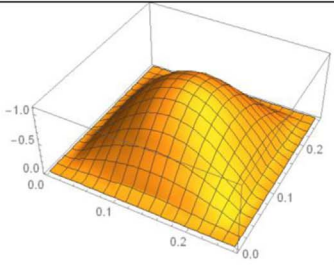
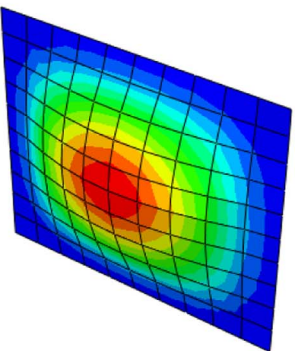
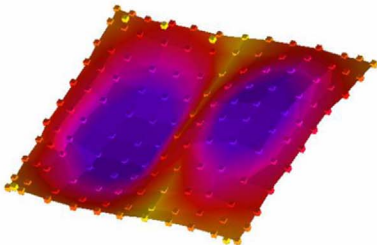
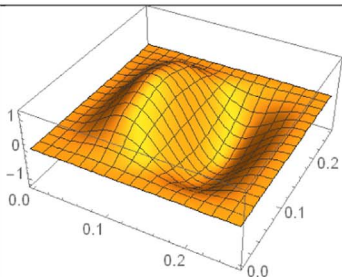
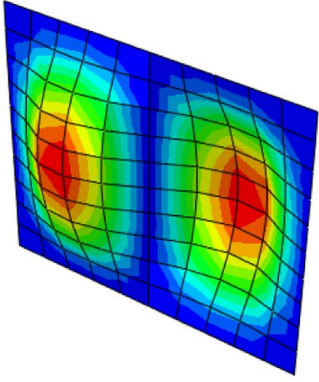
For the neoprene plate, the nonlinear response has been measured at five different force levels: 0.1N, 0.5N, 1.5N, 2.5N and 3.5N. The response has been measured at (0.12 m, 0.14 m), which is close to the centre, with the excitation located at (0.22 m, 0.24 m), which is close to a corner; this is the reason for the higher forces necessary with respect to the silicone plate. As the measurement has not been carried out at the centre of the plate, the response of the plate around the resonance of the fundamental mode (1,1) presents a contribution of the second mode

Table 1
Characteristics of the different models.

Sample	Mode shape m, n	Experimental Frequency Hz	Nonlinear Model				Linear Model				FEM	
			DOFs	Natural Freq Hz	N_x^s, N_y^s N/m	k N/rad	DOFs	Natural Freq Hz	N_x^s, N_y^s N/m	k N/rad	Natural Freq Hz	N_x^s, N_y^s N/m
Silicone	1, 1	21.00	39	20.90	105, 75	1000	507	21.29	115, 85	1000	21.08	115, 84
Neoprene	1, 1	13.16	54	13.14	40, 30	1000	507	13.43	50, 38	1000	13.07	50, 40
	2, 1	23.34		22.30				22.46			22.41	

Table 2

Mode shapes determined by experiments, linear model and FEM for the neoprene plate.

Mode (m, n)	Experiments	Linear Model	FEM
1, 1			
2, 1			

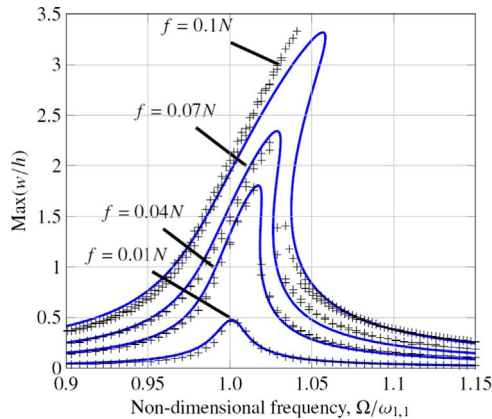


Fig. 8. Comparison between experimental and simulation (Kelvin Voigt model) results for the silicone plate; +, experimental results; —, numerical results.

(2,1). So, a larger model with 54 DOFs (sufficient to capture the first two resonances) has been used to get the nonlinear response during the numerical simulation. The time response of the two predominant modes, $w_{1,1}(t)$ and $w_{2,1}(t)$, at $\frac{\Omega}{\omega_{1,1}} = 0.6$ and at $\frac{\Omega}{\omega_{1,1}} = 1.6$ are shown in Figs. 9 and 10, respectively, for one excitation period. From the time response, it is clear that the displacements of both modes, at the point of measure, are out of phase (180° degree phase difference) before and in-phase (0° degree phase difference) after the first natural frequency. So, the total response has been calculated at each frequency while taking into account the phase difference between them. The frequency

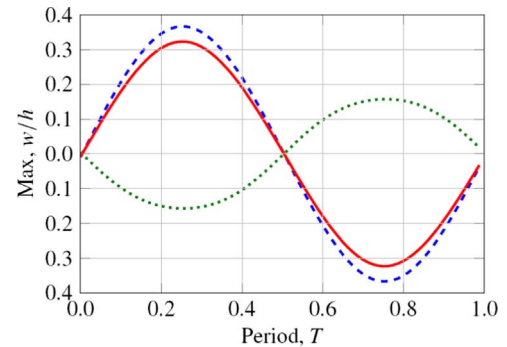


Fig. 9. Time response at (0.12 m, 0.14 m) for the neoprene plate for $\Omega/\omega_{1,1} = 0.6$; - -, $w_{1,1}(t)$ only; ···, $w_{2,1}(t)$ only; —, total response with contribution of both modes.

response calculated for the 2.5 N force level is shown in Fig. 11.

The comparison between the experimental and numerical results obtained from the 54 DOF model for the neoprene plate is shown in Fig. 12. The plate responses at the force levels of 0.1 and 0.5 N are practically linear and, as expected, they match very well with the experimental results. The retardation time η at these force levels is the same: 0.000924 s. As the force level is increased, the plate response becomes nonlinear with hardening behaviour. At the force levels of 1.5 N, 2.5 N and 3.5 N, the experimental results compare well with the numerical results. Moreover, the influence of second mode is also captured very well by the numerical model, as it can be seen by the good agreement of experimental and numerical results after the vibration peak, where the influence of the second mode is more relevant.

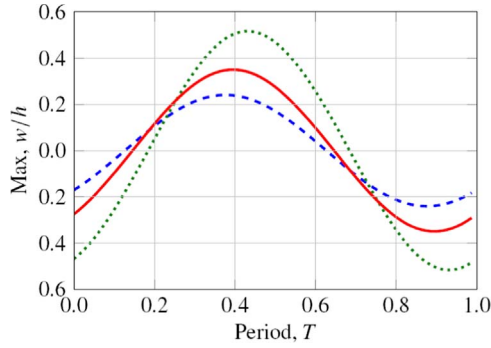


Fig. 10. Time response at (0.12 m, 0.14 m) for the neoprene plate for $\Omega/\omega_{1,1} = 1.6$; - - -, $w_{1,1}(t)$ only; - · -, $w_{2,1}(t)$ only; —, total response with contribution of both modes.

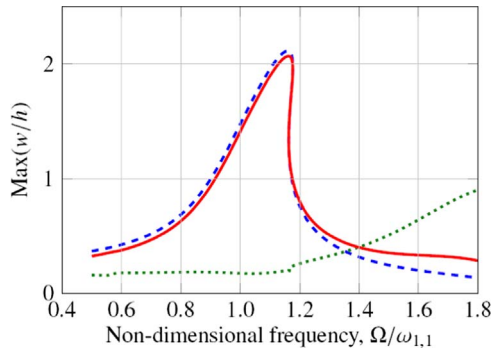


Fig. 11. Frequency response at (0.12 m, 0.14 m) for the neoprene plate; - - -, $w_{1,1}(t)$ only; - · -, $w_{2,1}(t)$ only; —, total response with contribution of both modes.

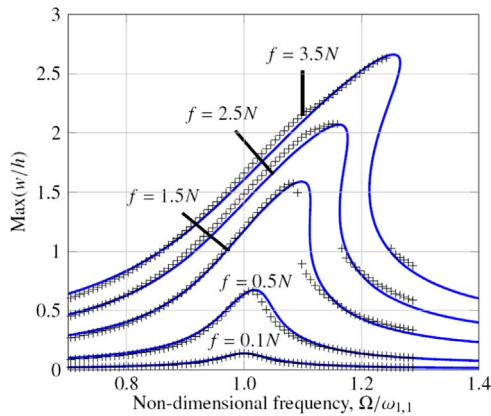


Fig. 12. Comparison between experimental and simulation (Kelvin Voigt model) results for the neoprene plate; +, experimental results; —, numerical results.

The small differences observed between experimental and numerical results could be due to the reduced number of DOFs used in the numerical model. Increasing the excitation levels, similar to silicone plate, it is observed that the retardation time η increases. The values corresponding to the five force levels in increasing order are 0.000924, 0.001016, 0.00115 and 0.00122 s. The retardation times η used in the numerical simulations to match the experimental data are non-dimensionalized, by divided by the initial retardation time η_{ini} obtained for the lowest force excitation (linear), and then plotted in Fig. 13 against the non-dimensionalized peak vibration amplitude. The graph shows the variation of the retardation time versus the vibration amplitude. This is quite interesting since a nonlinear damping model is

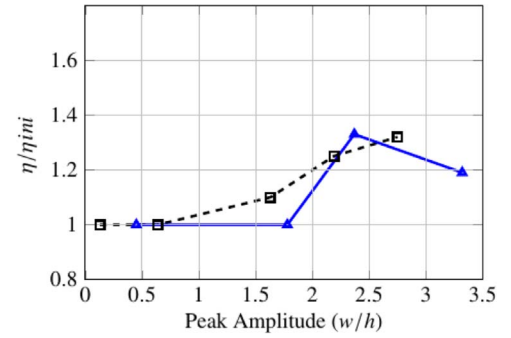


Fig. 13. Variation of retardation time with the peak vibration amplitude; —▲—, silicone plate; - -□-, neoprene plate.

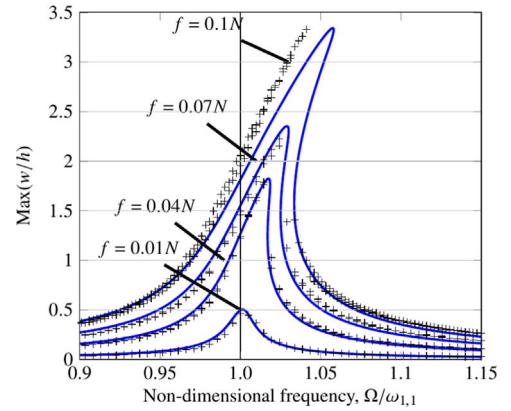


Fig. 14. Comparison between experimental and simulation (viscous damping model) results for the silicone plate; +, experimental results; —, numerical results.

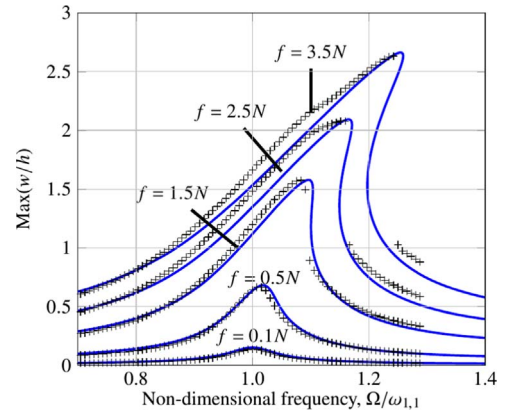


Fig. 15. Comparison between experimental and simulation (viscous damping model) results for the neoprene plate; +, experimental results; —, numerical results.

introduced by the Kelvin-Voigt viscoelasticity. Even the nonlinearity introduced by this model is unable to predict the variation of damping during large amplitude vibrations of both rubber plates. Hence retardation time has to be increased to match the experimental response at higher vibration amplitudes for both cases. Very interesting is that, for two plates of different rubber material and thickness, the same behaviour is obtained in Fig. 13, with an increase of the retardation time of about 20% and 30% for the silicone and neoprene plates respectively.

5.2.2. Comparison with the viscous damping model

The equivalent viscous damping model was also used to study the response of both the plates as per Eq. (23). The modal damping ratio $\zeta_{1,1}$ was varied at each force level for both plates to match the experimental

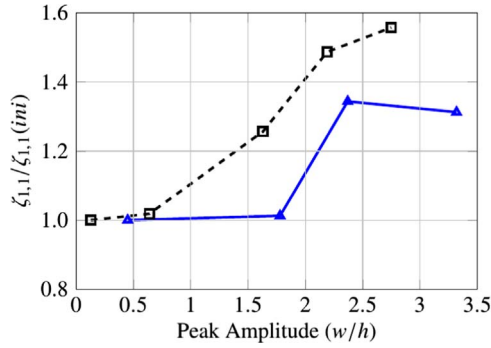


Fig. 16. Variation of the viscous damping with the peak vibration amplitude; —▲—, silicone plate; - -□- -, neoprene plate.

results. The comparison between the viscous damping model and experimental results for silicone and neoprene plates is shown in Figs. 14 and 15, respectively. The agreement is good also for this model, even if a little less good than for the viscoelastic dissipation. The modal damping ratio used in the simulation increases with the excitation level and vibration amplitude for both the plates. The damping ratios of the silicone plate are 0.008, 0.0081, 0.011 and 0.0105 for the force levels in increasing order. Similarly, the damping ratios of neoprene plate are 0.0382, 0.0389, 0.0465, 0.0558 and 0.0573 for the force levels in an increasing order.

The damping ratio $\zeta_{1,1}$, non-dimensionalized with respect to the damping value for linear vibrations $\zeta_{1,1}(ini)$, is plotted in Fig. 16 against the non-dimensionalized maximum vibration amplitude. Fig. 16 shows that in case of viscous damping, which is a linear dissipation model, an increase of damping ratio is identified by the nonlinear results, however the increase is more than the increase of retardation time (nonlinear damping model). The damping increase in this case with viscous damping model is different for the two plates of different rubber material. For the silicone plate, the damping ratio is basically constant up to peak vibration amplitudes about 1.8 times the thickness, and then it increases of about 30%. Instead the neoprene plate shows a regular increase of the damping ratio with the peak vibration amplitude, up to almost 60%.

Appendix A. Potential energy of a pre-loaded plate

The elastic potential energy of the vibrating plate, subjected to in-plane static pre-load in x and y direction uniform along each edge, is given by

$$U = \frac{1}{2} \int_{-h/2}^{h/2} \int_0^a \int_0^b (\sigma_x^t \epsilon_x + \sigma_y^t \epsilon_y + \tau_{xy}^t \gamma_{xy}) d z d x d y, \quad (A1)$$

where the total stresses (indicated with superscript t) are given by the contribution of static pre-load (indicated with superscript s) and the dynamic component due to vibration

$$\sigma_x^t = \sigma_x^s + \sigma_x, \quad \sigma_y^t = \sigma_y^s + \sigma_y, \quad \tau_{xy}^t = \tau_{xy}^s + \tau_{xy}, \quad (A2)$$

where it assumed that the static pre-load gives $\tau_{xy}^s = 0$. In Eq. (A1), the strains are only due to the vibration since it is assumed that the initial configuration of the plate, which is the reference configuration in the present study, is pre-loaded. Eq. (A2) are then substituted in Eq. (A1)

$$\begin{aligned} U &= \frac{1}{2} \int_{-h/2}^{h/2} \int_0^a \int_0^b [(\sigma_x^s \epsilon_x + \sigma_y^s \epsilon_y) + (\sigma_x \epsilon_x + \sigma_y \epsilon_y + \tau_{xy} \gamma_{xy})] d z d x d y \\ &= U_E + \frac{1}{2} \int_{-h/2}^{h/2} \int_0^a \int_0^b (\sigma_x^s \epsilon_x + \sigma_y^s \epsilon_y) d z d x d y, \end{aligned} \quad (A3)$$

where U_E is given in Eq. (7). Substituting Eqs. (1a,b) into Eq. (A3), the following expression is obtained for the elastic potential energy of the plate

$$U = U_E + \frac{1}{2} \int_{-h/2}^{h/2} \int_0^a \int_0^b (\sigma_x^s \epsilon_{x,0} + \sigma_y^s \epsilon_{y,0} + \sigma_x^s z k_x + \sigma_y^s z k_y) d z d x d y = U_E + \frac{1}{2} \int_0^a \int_0^b (N_x^s \epsilon_{x,0} + N_y^s \epsilon_{y,0}) d x d y, \quad (A4)$$

where the assumption of uniform σ_x^s and σ_y^s through the thickness is used. The integral on the right inside of Eq. (A4) gives the plate potential energy due to initial in-plane static pre-load.

6. Conclusions

Geometrically nonlinear vibrations of clamped rubber plates have been studied experimentally and theoretically. Two thin rectangular plates made of different materials and thicknesses have been considered; a silicone rubber plate with 0.0015 m of thickness and a neoprene rubber plate with 0.003 m of thickness. The viscoelastic characteristics of the material have been modelled using the Kelvin-Voigt model and the geometrically nonlinear strain-displacement relationships using the von Kármán nonlinear plate theory. Numerical results are matching very well the experimental responses, and allow to identify the damping parameters. The increasing trend of retardation time with increasing vibration amplitude has been found in both the plates. This behaviour is similar to the one observed for the equivalent viscous damping model but the increase in retardation time is lesser. The difference in the behaviour is surely due to the nonlinear damping terms introduced by the Kelvin-Voigt viscoelastic dissipation model, since the linear part of the two dissipation models can be made equivalent. It must be pointed out here that the plates investigated are hardening type systems, for which the nonlinear damping terms are introducing an increase in the dissipation. However, this increase is not sufficient to follow the experimental behaviour, so that the retardation time must be increased with the vibration amplitude, i.e. just as a global increase of damping ratio found with the viscous damping model. At this point, it would be interesting to investigate a softening type system with rubber shells, since in that case the nonlinear damping terms should reduce the dissipation with the increase of the vibration amplitude. This is left for a future study. However, what seems clear from this study is that traditional damping models, namely viscous damping and Kevin-Voight viscoelasticity, present a poor description of the dissipation in rubber plates. In fact, dissipation parameter must be varied with the force excitation (or the peak vibration amplitude) in order to reproduce correctly the vibration response experimentally measured.

Acknowledgements

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